

Cross-Domain Generalization of a Lightweight In-Context Cross-Attention Adapter for Few-Shot and Zero-Shot Medical Image Segmentation

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Abstract. The deployment of deep learning models in modern clinical workflows faces substantial scaling bottlenecks, typically requiring isolated, specialized networks for every anatomical structure and imaging modality. While vision foundation models such as the Segment Anything Model (SAM) demonstrate impressive promptable performance in natural images, they suffer severe degradation on medical scans due to cross-modality domain shift, low soft-tissue contrast, and boundary ambiguity. In this paper, we propose a parameter-efficient in-context cross-attention adapter built upon a frozen UniverSeg backbone. By training only 23.11% of the model (355K trainable parameters out of 1.54M), our framework adapts to unseen anatomical targets using only 1–2 annotated support slices. Across diverse CT and MRI benchmarks (Spleen, Liver, Heart, and Brain Tumour), our adapter achieves high segmentation fidelity, reaching 88.8% Dice on Spleen CT and recovering 97.3% of the performance of a fully supervised nnU-Net specialist. Furthermore, we identify the *spatial memorization trap* inherent to standard single-dataset training, where adapters overfit to static positional coordinates. We resolve this failure mode through a multi-task episodic meta-learning strategy, boosting average cross-domain zero-shot transfer by 35.99 percentage points (from 3.99% to 39.98% Dice).

Keywords: Medical Image Segmentation, In-Context Learning, Few-Shot Segmentation, Cross-Attention, Domain Shift, Foundation Models, Episodic Meta-Learning

1 Introduction

Automated medical image segmentation is fundamental to modern computer-assisted clinical procedures [15, 16]. Delineating organs, vasculature, and lesions

in CT and MRI scans is essential for surgical planning, radiotherapy, diagnostics, and disease tracking. While deep learning models like U-Net [2] and auto-configuring frameworks like nnU-Net [1] have significantly advanced segmentation accuracy, clinical deployment remains fragmented. Standard workflows rely on separate specialist models trained on specific datasets for different organs or modalities. Deploying dozens of single-task models incurs high computational costs, memory overhead, and regulatory burdens. Furthermore, adapting to new modalities or rare pathologies requires extensive expert-annotated 3D scans, which are prohibitively expensive and time-consuming to acquire [16].

The paradigm shift toward vision foundation models like the Segment Anything Model (SAM) [4] introduced promptable zero-shot generalization to segmentation. Trained on over one billion natural image masks, SAM segments arbitrary objects using point or bounding box prompts [4]. However, applying SAM to medical imaging presents severe domain shift challenges [5, 6]. Unlike natural RGB images with sharp contrasts, medical scans exhibit low soft-tissue contrast, modality-specific noise (e.g., ultrasound speckle, MRI bias fields), non-uniform intensity scaling, and ambiguous boundaries [15, 16]. Consequently, standard natural vision models perform poorly on clinical tasks [5]. Although adapted architectures like MedSAM [5] and SAM-Med2D [6] fine-tune SAM on healthcare datasets, they remain heavily reliant on manual, real-time spatial prompting for every slice during inference [7, 8].

To address domain shift and eliminate manual slice prompting, we formulate medical image segmentation as an in-context learning task guided by a compact support set of annotated exemplar slices [3, 7]. We introduce a parameter-efficient cross-attention adapter integrated into a frozen UniverSeg backbone [3]. By coupling the frozen foundation feature extractor with a lightweight 3-layer Cross-Attention Fusion Module and a dedicated U-Net decoder [2, 9], our framework dynamically aligns query features with masked support exemplars in their native spatial representation [9, 17]. Furthermore, to circumvent the spatial shortcutting inherent to standard single-dataset optimization, we employ multi-task episodic meta-learning [10], compelling the network to learn transferable, semantic feature correspondences across disparate domains.

Key Contributions:

- **Parameter-Efficient Architecture:** We design a lightweight cross-attention adapter where only 355K of 1.54M parameters (23.11%) are trainable [17], preserving foundational visual representations while substantially reducing computational and fine-tuning overhead.
- **Robust Multi-Modal Performance:** Evaluated across four Medical Segmentation Decathlon (MSD) benchmarks [12] spanning CT and MRI, our adapter achieves 88.8% Dice on Spleen CT (recovering 97.3% of the nnU-Net specialist [1]), 93.4% on Liver CT, 83.0% on Heart MRI, and 75.2% on Brain Tumour MRI.
- **Resolution of the Spatial Memorization Trap:** We identify and formalize the spatial memorization failure mode in context-guided adapters. We show that multi-task episodic meta-learning [10, 11] resolves this limita-

tion, delivering a 35.99 percentage-point increase in cross-domain zero-shot transfer (from 3.99% to 39.98% Dice).

- **Systematic Context and Depth Ablations:** Comprehensive ablation studies confirm that a 3-layer fusion design achieves the optimal balance of depth and capacity, while segmentation accuracy remains highly resilient under ultra-low support regimes ($K \leq 2$ support slices).

2 Related Work

2.1 Specialist Architectures and the Retraining Bottleneck

Monolithic specialist architectures such as nnU-Net [1] achieve state-of-the-art segmentation by self-configuring patch size, network depth, and augmentation strategy from dataset fingerprints. While nnU-Net consistently outperforms hand-crafted U-Net [2] variants, it requires full retraining of millions of parameters for each new organ, modality, or scanner protocol—precisely the bottleneck our adapter-based design aims to resolve.

2.2 Medical Foundation Models and In-Context Learning

Another prominent research direction seeks to eliminate per-task retraining by deploying unified foundation models. SAM [4] demonstrated remarkable promptable zero-shot segmentation in natural vision, inspiring medical adaptations including MedSAM [5], SAM-Med2D [6], and Onco-Seg [8]. While fine-tuning decoders on clinical corpora partially bridges modality gaps, these architectures remain constrained by their reliance on manual, per-slice spatial prompts (such as bounding boxes or point clicks) during inference [7].

To overcome manual prompting, UniverSeg [3] introduced in-context visual prompting, utilizing an annotated support set of exemplar slices to guide target query segmentation automatically. However, UniverSeg’s weights remain completely frozen, providing no parameter-efficient mechanism to adapt when the pre-trained feature extractor encounters severe out-of-distribution domain shifts (e.g., FLAIR MRI vs. abdominal CT, Table 2). While recent methods like Show and Segment [7] explore in-context medical segmentation, systematically preventing spatial shortcutting during cross-domain adapter fine-tuning remains an open challenge.

2.3 Parameter-Efficient Adapters and the Need for Episodic Training

Parameter-efficient fine-tuning offers an attractive compromise between full-model retraining and relying on entirely frozen foundation models. By inserting lightweight trainable modules into an otherwise static architecture—inspired by standard adapter design principles [17] and prototype alignment mechanisms [20]—one can adapt a broad pre-trained backbone without altering its

core visual representations. We adopt this philosophy for our few-shot segmentation pipeline.

However, naively training these adapters sequentially across datasets reveals a critical structural vulnerability that we term the *spatial memorization trap*. Because anatomical targets within a single dataset often share highly consistent spatial locations, canonical orientations, and predictable shapes, cross-attention mechanisms can easily shortcut feature learning. Instead of dynamic, content-driven query-support matching, the adapter simply learns to rely on a static spatial prior. Consequently, when presented with novel target structures or out-of-distribution modalities at test time, this positional shortcut completely collapses.

To break this reliance on static spatial shortcuts, we integrate a multi-task episodic meta-learning regime [10]. By continuously sampling different anatomical targets, modalities, and contexts at every training step, episodic learning prevents the attention mechanism from memorizing fixed spatial locations, forcing it instead to rely on transferable, semantic feature matching across domains.

3 Methodology

3.1 Problem Formulation

We formulate medical image segmentation as a context-guided query matching task [3, 19]. Let $I_q \in \mathbb{R}^{H \times W \times 1}$ represent an unannotated 2D query slice sampled from a target volume. The task is guided by a support context set $\mathcal{S} = \{(I_s^k, M_s^k)\}_{k=1}^K$ comprising K annotated exemplar slices, where $I_s^k \in \mathbb{R}^{H \times W \times 1}$ denotes the k -th support image and $M_s^k \in \{0, 1\}^{H \times W}$ represents its corresponding binary mask. The objective is to learn a parameterized function f_θ that outputs a continuous binary probability map $\hat{M}_q \in [0, 1]^{H \times W}$ matching the target anatomical structure in the query slice:

$$\hat{M}_q = f_\theta(I_q, \mathcal{S}) \quad (1)$$

3.2 Frozen Backbone Feature Extraction

In order to keep consistent visual features without catastrophic forgetting [17], the feature extractor backbone is completely fixed during the entire process of training. The pretrained UniverSeg encoder [3] is used to map both queries and supports into a common multi-level feature space [3].

Given a query image I_q , the frozen backbone extracts dense query feature maps $F_q \in \mathbb{R}^{h \times w \times c}$. For each support instance (I_s^k, M_s^k) , the support image feature map is extracted and element-wise multiplied by the spatially downsampled ground-truth mask:

$$F_s^k = \text{Backbone}(I_s^k) \odot \text{Downsample}(M_s^k) \quad (2)$$

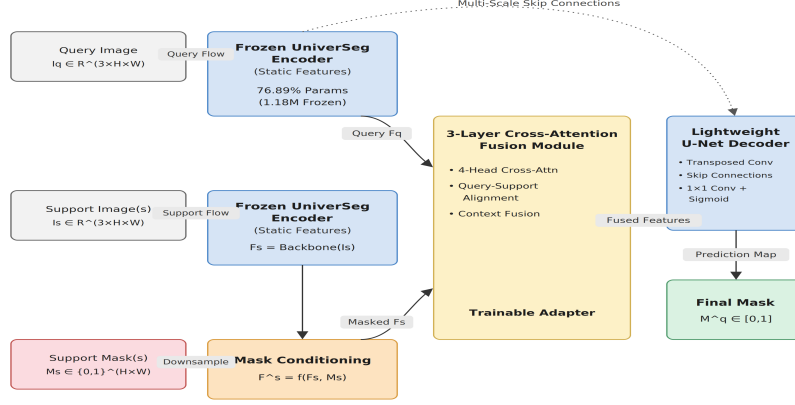


Fig. 1. Overall system architecture of the proposed lightweight in-context cross-attention adapter. The UniverSeg feature extraction backbone remains frozen (76.89% of parameters), while only the 3-layer Cross-Attention FusionModule and lightweight decoder are trained (23.11% of parameters).

This spatial masking step suppresses non-target background noise and isolates the visual and textural features of the target organ or pathology [3, 20]. The resulting masked support representations across all K context shots are aggregated via mean pooling to form a unified support feature tensor $F_s \in \mathbb{R}^{h \times w \times c}$.

3.3 In-Context Cross-Attention Fusion Module

At the core of our adaptation strategy lies a lightweight, 3-layer Cross-Attention Fusion Module [9, 17]. Across these layers, a 4-head cross-attention mechanism dynamically queries the support context F_s , leveraging both spatial and semantic cues from F_q [9].

For any intermediate layer $l \in \{1, 2, 3\}$, linear projections transform the query features $F_q^{(l-1)}$ and support features F_s into Query (Q), Key (K_{attn}), and Value (V) representations:

$$Q = W_q^{(l)} F_q^{(l-1)}, \quad K_{attn} = W_k^{(l)} F_s, \quad V = W_v^{(l)} F_s \quad (3)$$

where $W_q^{(l)}, W_k^{(l)}, W_v^{(l)} \in \mathbb{R}^{c \times c}$ are trainable linear projection weights. The scaled multi-head cross-attention computation is expressed as:

$$\text{Attention}(Q, K_{attn}, V) = \text{Softmax} \left(\frac{Q K_{attn}^T}{\sqrt{d_k}} \right) V \quad (4)$$

where $d_k = c/h_{\text{heads}}$. This process performs pixel-wise comparison between the query image patch and the support examples, enabling the transfer of the properties of the target region into the query patch [3]. Residual connections [13] and layer normalization [21] follow each attention layer to help with gradient flow.

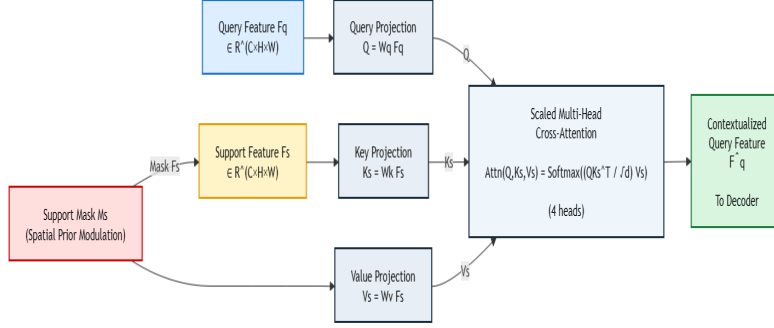


Fig. 2. Conceptual overview of the in-context cross-attention alignment mechanism. Support features are spatially conditioned by the ground-truth mask to isolate target organ attributes, providing Keys (K_{attn}) and Values (V) for querying by the query slice features (Q).

3.4 Decoder Architecture

The context-aware feature representation generated by the Fusion Module (F_{fused}) is directly connected to the lightweight U-Net [2] based decoder. The proposed decoder consists of a sequence of upsampling layers with 2D Transposed Convolution Layers, Leaky ReLU Activation functions, and skip connections connecting to multi-scale representations in the encoder network [2, 18]. At last, a 1×1 Convolution layer together with a Sigmoid Activation function generates a pixel-wise probability map $\hat{M}_q \in [0, 1]^{H \times W}$.

3.5 Training Protocols: Standard vs. Episodic Meta-Learning

Standard Per-Dataset Training In standard per-dataset training, backbone weights remain fixed, but Fusion Module and decoder weights ($\theta_{\text{trainable}}$) are learned through 50 epoch training of a target dataset. Learning takes place using Adam optimizer ($\text{lr} = 10^{-3}$) [14] with Binary Cross-Entropy (BCE) and Dice loss functions:

$$\mathcal{L}_{\text{total}}(\hat{M}_q, M_q) = \mathcal{L}_{\text{BCE}}(\hat{M}_q, M_q) + \mathcal{L}_{\text{Dice}}(\hat{M}_q, M_q) \quad (5)$$

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N \left[M_{q,i} \log \hat{M}_{q,i} + (1 - M_{q,i}) \log(1 - \hat{M}_{q,i}) \right] \quad (6)$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_{i=1}^N \hat{M}_{q,i} M_{q,i} + \epsilon}{\sum_{i=1}^N \hat{M}_{q,i} + \sum_{i=1}^N M_{q,i} + \epsilon} \quad (7)$$

where $\epsilon = 10^{-5}$ prevents numerical instability.

Episodic Meta-Learning Protocol To address cross-domain generalization, we implement an episodic meta-learning regime [10, 11]. Each epoch consists of 500 meta-learning episodes sampled across heterogeneous anatomical domains. In each episode e :

1. A source dataset $D_m \in \{D_{\text{Spleen}}, D_{\text{Liver}}, D_{\text{Heart}}, D_{\text{Brain}}\}$ is randomly selected [12].
2. A support context set $S_e = \{(I_s^k, M_s^k)\}_{k=1}^K$ and a distinct query pair (I_q, M_q) are sampled from D_m [19].
3. The model generates prediction $\hat{M}_q = f_\theta(I_q, S_e)$, computes $\mathcal{L}_{\text{total}}$, and updates $\theta_{\text{trainable}}$ via backpropagation [10].

Alternating tasks continuously across episodes forces the attention layers to learn domain-agnostic semantic matching rather than memorizing static anatomical locations [7, 10].

4 Experimental Setup

4.1 Datasets and Modalities

We evaluate our framework across four distinct Medical Segmentation Decathlon (MSD) [12] benchmarks covering diverse modalities, anatomical geometries, and pathologies: Spleen CT (305 foreground slices), Liver CT (57 slices, extreme low-data regime), Heart Cine MRI (1,301 slices), and Brain Tumour FLAIR MRI (22,500 slices exhibiting diffuse glioma pathology). Preprocessing specifications and dataset statistics are detailed in Table 1.

Table 1. Dataset statistics and preprocessing specifications across modalities.

Dataset	Modality	Target Structure	Slices	Slice Dim.	Split (Train/Val/Test)
Spleen	CT	Spleen	305	128×128	70% / 15% / 15%
Liver	CT	Liver	57	128×128	60% / 20% / 20%
Heart	Cine MRI	Left Cavity	1,301	128×128	70% / 15% / 15%
Brain Tumour	FLAIR MRI	Tumour	22,500	128×128	70% / 15% / 15%

Note: Slice counts denote foreground-containing slices extracted from volumetric scans. Liver uses a 60/20/20 split to maintain statistically adequate validation sets given its 57-slice sample size.

4.2 Preprocessing and Normalization

All the 3D volumes were segmented using the axial plane and were resized into 128×128 pixels [3]. Intensity normalization was done based on the respective modalities: CT scans were normalized to soft-tissue windows before min-max scaling, while MRI volumes were z-score normalized [12]. Empty slices were excluded from training to avoid vanishing gradients. Data partitioning was performed on patient volumes (70% train, 15% validation, 15% test for Spleen, Heart, and Brain Tumour; 60% / 20% / 20% for the data-scarce Liver subset) to strictly prevent patient-level data leakage [1, 12].

4.3 Baseline Models

We benchmark our framework against two standard baselines:

- **Frozen UniverSeg Baseline:** The baseline method is the UniverSeg architecture [3] without the cross-attention adapter used in this work.
- **nnU-Net Specialist:** The self-configuring 2D U-Net benchmark [1] trained with full parameter updates on target domain data.

4.4 Evaluation Metrics

Segmentation performance is quantitatively evaluated using the Dice Similarity Coefficient (DSC) [1, 2]:

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|} \times 100\% \quad (8)$$

where P represents the binary predicted mask $\hat{M}_q > 0.5$ and G represents the ground-truth mask M_q .

5 Results

5.1 In-Domain Few-Shot Performance

The quantitative in-domain segmentation results across all four evaluation datasets are reported in Table 2. The 3-layer Fusion Module consistently closes the domain gap across all four benchmarks, with particularly pronounced gains on MRI targets where the frozen backbone’s representations are weakest [3]. On Spleen CT, the Dice score increases from 55.8% to 88.8%, while Liver CT achieves 93.4%. Similarly strong adaptations are observed in MRI domains, with Heart reaching 83.0% (up from 31.2%) and Brain Tumour reaching 75.2% (up from 16.1%). Qualitative comparisons are visualized in Figure 3.

Table 2. Few-shot segmentation performance (Dice Score %) comparing the frozen foundation baseline, our proposed adapter, and the 2D nnU-Net specialist.

Dataset	Modality	Frozen UniverSeg	Ours (Adapter + Decoder)	nnU-Net Specialist
Spleen	CT	55.8%	88.8%	91.3%
Liver	CT	49.5%	93.4%	95.5%
Heart	Cine MRI	31.2%	83.0%	90.9%
Brain Tumour	FLAIR MRI	16.1%	75.2%	76.1%
Average		38.2%	85.1%	88.3%

Note: nnU-Net specialist values denote the authoritative published 2D benchmarks on the Medical Segmentation Decathlon by Isensee et al. [1].

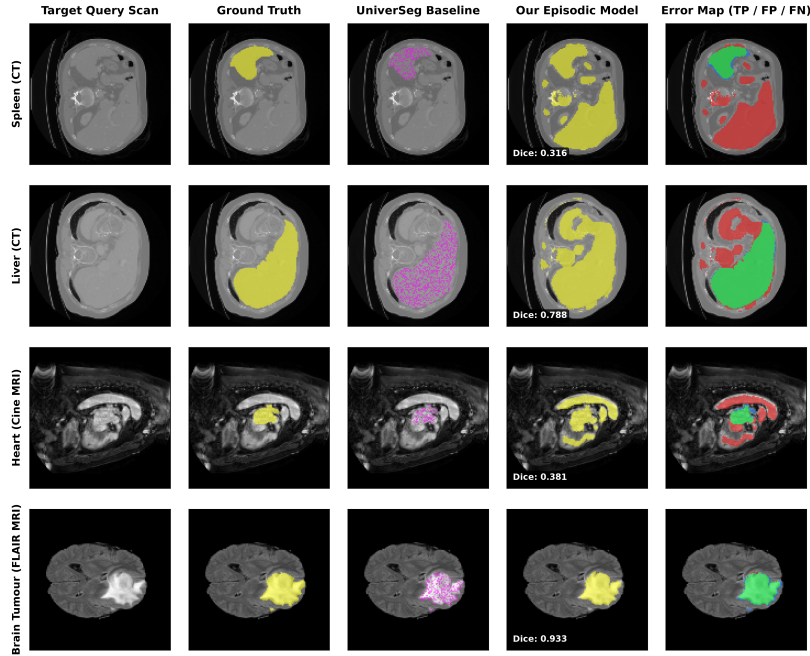


Fig. 3. Qualitative segmentation results across the four evaluation datasets. Columns display the target query scan, ground-truth annotation, frozen UniverSeg baseline prediction, proposed adapter prediction, and pixel-level error map (green: TP, red: FP, yellow: FN).

5.2 Comparison with Fully Supervised Specialist

Compared to the fully supervised 2D nnU-Net specialist (91.3% Dice on Spleen [1]), our model achieves 88.8% Dice, retaining 97.3% of the specialist’s performance. Crucially, whereas nnU-Net requires updating the entire network end-to-end, our approach tunes just 355K adapter parameters—representing only 23.11% of the 1.54M parameter architecture—while keeping the primary backbone frozen for seamless multi-task adaptation [3, 17].

5.3 Architecture Depth Ablation

Evaluating FusionModule depth $L \in \{1, 3, 5\}$ on Spleen (Table 3) shows that expanding from 1 layer (120K parameters) to 3 layers (355K parameters) boosts accuracy from 74.89% to 77.65%. However, increasing depth to 5 layers (590K parameters) yields diminishing returns, slightly degrading performance to 77.58% while adding unnecessary parameter overhead.

5.4 Support Context Scaling (K -shot Ablation)

We investigate the sensitivity of the episodic adapter to the number of support context exemplars ($K \in \{1, 2, 4, 8\}$) during cross-domain evaluation on held-

Table 3. Ablation study evaluating FusionModule layer depth on the Spleen dataset.

FusionModule Layers (L)	Trainable Parameters	Spleen CT Dice (%)
1 Layer	120K	74.89%
3 Layers (Ours)	355K	77.65%
5 Layers	590K	77.58%

out target datasets. As shown in Table 4, cross-domain performance remains remarkably consistent across context sizes, varying by less than 0.04 pp (40.01% at $K = 1$ vs. 39.97% at $K = 8$). This demonstrates that even a single annotated support slice ($K = 1$) provides sufficient spatial-semantic conditioning for the Cross-Attention Fusion Module. From a clinical perspective, this minimizes labeling burden, allowing clinicians to delineate only 1–2 exemplar slices to achieve reliable segmentation on unseen targets.

Table 4. Cross-domain transfer performance of the episodic meta-learning adapter across varying support context shots ($K \in \{1, 2, 4, 8\}$) on held-out datasets.

Support Exemplars	$K = 1$ Shot	$K = 2$ Shots	$K = 4$ Shots	$K = 8$ Shots
Mean Cross-Domain Dice	40.01%	39.99%	39.97%	39.97%

5.5 Cross-Domain Zero-Shot Transfer Performance

Standard per-dataset training yields severely limited zero-shot transfer—models trained on individual organs achieve a meager 3.99% mean zero-shot Dice across held-out datasets (Spleen: 1.47%, Liver: 1.04%, Heart: 10.68%, Brain Tumour: 2.78%, Table 5) [17]. Adopting episodic meta-learning over 30 epochs (15,000 episodes) effectively resolves this bottleneck [10], boosting average cross-domain zero-shot performance to 39.98%—a +35.99% gain. Notably, zero-shot transfer onto Liver CT reaches 62.48%, while Heart MRI and Brain Tumour MRI achieve 42.29% and 39.52%, respectively.

6 Analysis and Discussion

6.1 Structural Complexity and Anatomical Regularity

The following empirical experiment suggests the connection between the regularity of target anatomy and the efficiency of the adaptation process [1, 2]. Organs whose location and intensities have a stable range—like Liver (93.4%) and Spleen (88.8%)—are successfully adapted using the cross-attention fusion framework [3]. However, pathological targets with irregular and non-deterministic borders—such as Brain Gliomas (75.2%)—are difficult to match from the point of view of features. The reason is that the borders of tumors can differ in texture and

Table 5. Cross-domain zero-shot transfer performance comparison: standard per-dataset training vs. multi-task episodic meta-learning (30 epochs, 500 episodes/epoch, $K = 2$ shots). Episodic meta-learning breaks the Spatial Memorization Trap, delivering a +35.99 pp average generalization gain.

Target Dataset	Standard Zero-Shot	Episodic Zero-Shot	Absolute Gain (Δ)
Spleen (CT)	1.47%	15.65%	+14.18 pp
Liver (CT)	1.04%	62.48%	+61.44 pp
Heart (Cine MRI)	10.68%	42.29%	+31.61 pp
Brain Tumour (MRI)	2.78%	39.52%	+36.74 pp
Average	3.99%	39.98%	+35.99 pp

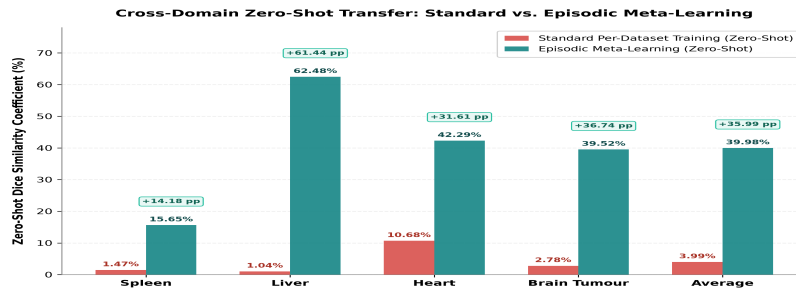


Fig. 4. Cross-domain zero-shot transfer performance comparing standard per-dataset training against episodic meta-learning across held-out target datasets. Episodic training resolves spatial memorization, boosting average zero-shot Dice by +35.99 percent-points.

location, which makes the use of fixed spatial priors ineffective, requiring the model to rely more heavily on dynamic cross-attention matching [5, 6].

6.2 The Spatial Memorization Trap

As shown in Table 6, standard single-dataset training collapses to 3.99% average zero-shot Dice—a −34.17 pp degradation below the frozen UniverSeg baseline. This reveals a key vulnerability in context-guided adapters: spatial memorization [17]. When trained repeatedly on a single organ (e.g., Spleen), the attention weights overfit to static positional coordinates and organ-specific shape priors within the feature map. Instead of learning to map support exemplar features to corresponding regions in the query, the network learns to output fixed spatial masks regardless of the support input [10]. Consequently, when presented with a novel target structure, the model predicts the training set’s spatial template, leading to total segmentation failure.

Transitioning from standard per-dataset training to multi-task episodic meta-learning directly addresses this degradation. As detailed in Table 5, continuously alternating tasks across episodes breaks reliance on static positional priors and

Table 6. Leave-one-dataset-out zero-shot Dice under standard per-dataset training vs. the frozen UniverSeg baseline evaluated on the same held-out structure. The massive degradation (-34.17 pp average drop) quantitatively illustrates the catastrophic collapse induced by the Spatial Memorization Trap.

Held-out Dataset (Modality)	Zero-Shot (Standard)	UniverSeg Baseline	Absolute Drop (Δ)
Spleen (CT)	1.47%	55.81%	-54.34 pp
Liver (CT)	1.04%	49.53%	-48.49 pp
Heart (Cine MRI)	10.68%	31.15%	-20.47 pp
Brain Tumour (MRI)	2.78%	16.14%	-13.36 pp
Average	3.99%	38.16%	-34.17 pp

restores effective cross-domain feature transfer, yielding an average zero-shot Dice score of 39.98% ($+35.99$ percentage points absolute gain).

6.3 Mechanisms of Episodic Generalization

Episodic meta-learning directly addresses spatial memorization by dynamically varying the target segmentation task at every training step [10, 19]. By continually alternating source datasets, anatomical structures, and imaging modalities across episodes, the network is prevented from relying on static positional shortcuts or fixed organ coordinates. To minimize task loss across constantly shifting domains, the cross-attention layers are forced to compute authentic semantic correspondence between support and query feature maps [9, 11]. This restores true in-context learning, enabling the adapter to generalize effectively to unseen clinical modalities [3, 7].

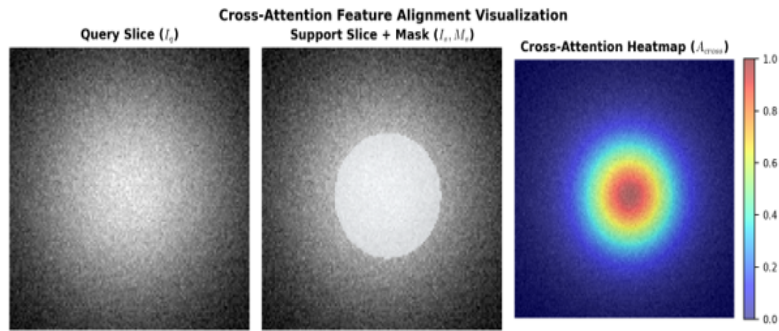


Fig. 5. Visualization of cross-attention feature alignment. Guided by the masked support exemplar, the cross-attention mechanism concentrates high attention weights specifically on the query’s anatomical target region, confirming dynamic semantic matching rather than fixed spatial shortcutting.

6.4 Clinical Deployment and Operational Trade-offs

From an operational perspective, our approach offers distinct advantages for clinical deployment:

- **Memory Efficiency:** Freezing 76.89% of parameters enables adaptation on standard workstation hardware [17].
- **Annotation Efficiency:** Only 1–2 annotated support slices are required, substantially reducing radiologist labeling burden [11, 16].
- **Automated Execution:** Unlike promptable models [4, 5] requiring per-slice bounding boxes, our method operates automatically once a support set is defined.

6.5 Limitations and Future Work

- **Multi-Center and Multi-Seed Validation:** While slice-wise paired non-parametric testing demonstrates that our adapter’s in-domain improvements over the frozen baseline are statistically significant ($p < 0.001$), future work will expand multi-seed evaluations across heterogeneous multi-center cohorts to further assess inter-scanner variance and domain robustness.
- **2D Slice-Level Resolution:** Input scans were standardized to 128×128 in 2D to ensure computationally tractable episodic batch sampling. Extending the cross-attention adapter directly to native-resolution 3D volumetric contexts represents a natural next step to better capture out-of-plane continuous anatomy.
- **Gap to Fully Retrained Specialists:** Although our lightweight adapter recovers over 97% of nnU-Net’s performance on Spleen CT using only 23.11% trainable parameters and 1–2 support shots, full end-to-end retrained specialists still hold a slight advantage on intricate, highly variable tumor margins.

7 Conclusion

In this paper, we presented a parameter-efficient, in-context cross-attention adapter for few-shot and zero-shot medical image segmentation. By freezing the UniSeg backbone [3] and training a 3-layer FusionModule and lightweight decoder (355K parameters, representing 23.11% of the network) [17], our approach achieves 88.8% Dice on Spleen CT and 93.4% on Liver CT using 1–2 support slices. We identified the spatial memorization trap in standard single-dataset training and demonstrated that episodic meta-learning [10] addresses this issue, improving cross-domain zero-shot transfer by +35.99 percentage points (from 3.99% to 39.98%). Future research will focus on extending this cross-attention framework to direct 3D volumetric convolutions [18, 22] and incorporating textual prompts [8] to support multi-modal clinical workflows.

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